Stat 486

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The Pursuit of Happiness

THE QUESTIONS

Which life factors are the strongest predictors of subjective happiness, and do these factors differ by sex?

Does the data reveal natural grouping among respondents based on wellbeing profiles?

AGENDA

01 Research Questions

02 Data

03 EDA

04 Supervised Models

05 Unsupervised Model

06 Conclusions / Questions

OUR DATA

The General Social Survey

GSS

The General Social Survey is “a series of nationally representative cross-sectional interviews in the United States that have occurred since 1972.”

Managed by NORC at the University of Chicago, it provides a massive, multi-dimensional dataset (thousands of variables) on complex social issues like gender, labor, and education.

Basically, sociological data about the US second only to the Census

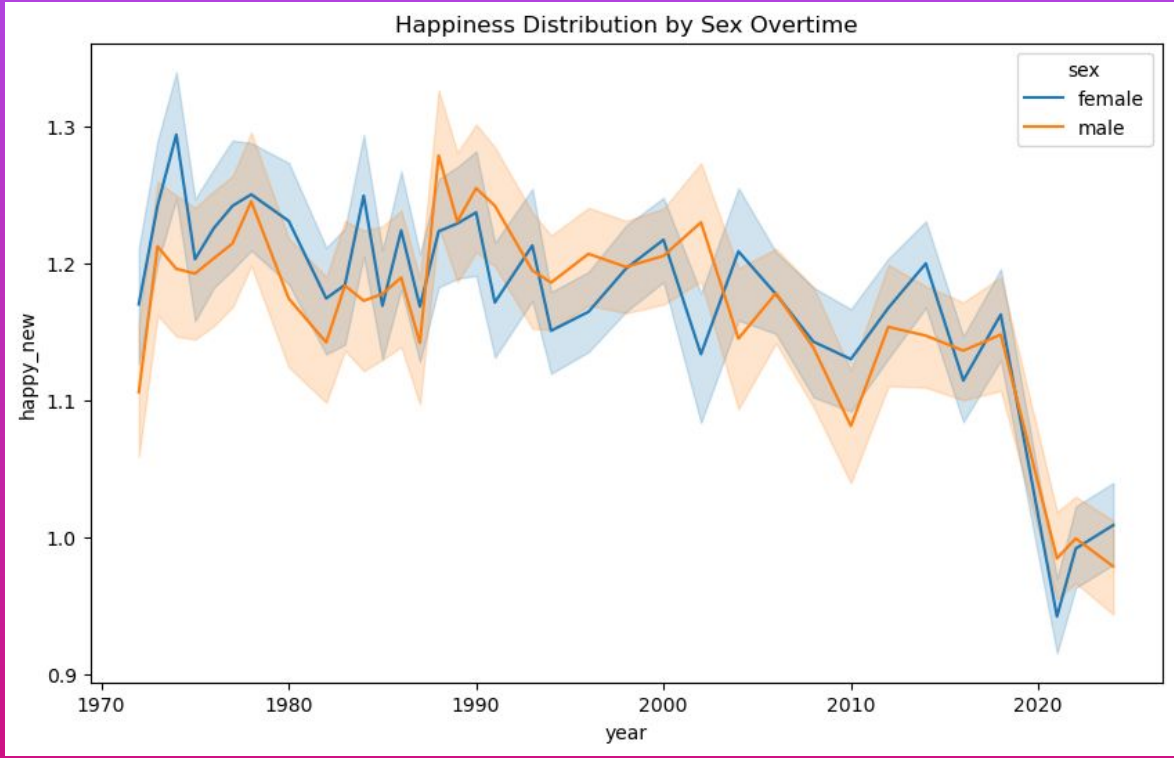
KEY VARIABLES

01	Target	happy — not too happy (0), pretty happy (1), very happy (2)
02	Features Used	health, marital happiness (hapmar), life excitement (life), age, education, religion, marital status, work status, sex, year
03	Encoding Decisions	ordinal encoding for ordered variables (health, life, happy), binary for sex, one-hot for nominal (race, marital status, religion)
04	Class Imbalance	55% pretty happy, 31% very happy, 14% not too happy → addressed with SMOTE (logistic regression) and class_weight='balanced' (random forest)

EDA

STEVENSON & WOLFERS

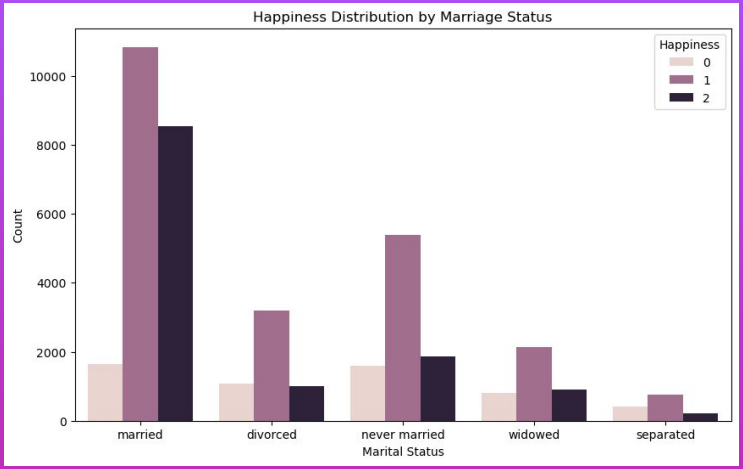
"PARADOX OF DECLINING FEMALE HAPPINESS"

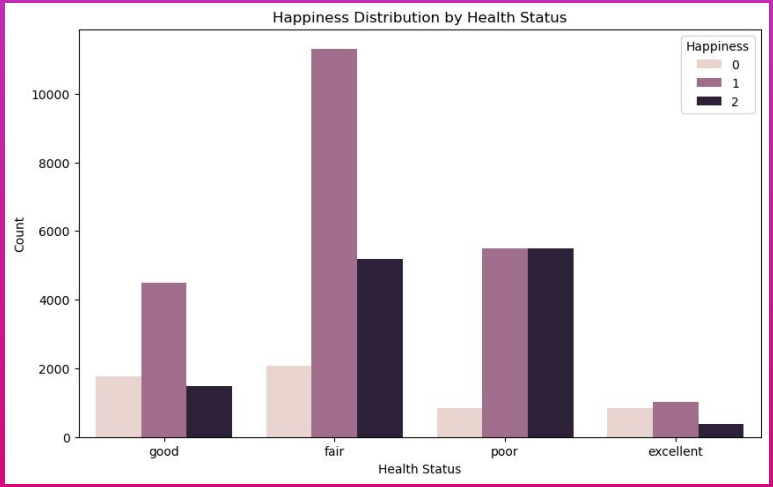


EDA

Happiness Distributions for:

- Marital Status
- Health Status





SUPERVISED MODELING APPROACH

Logistic Regression & Random Forest Classifier

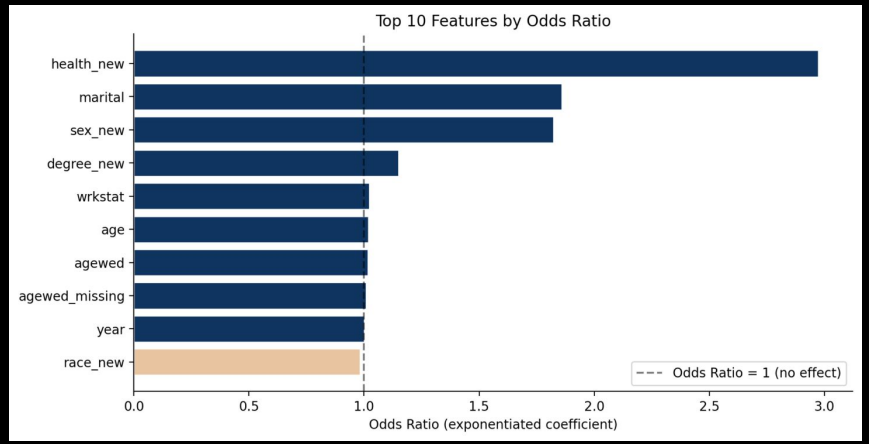
- → Two models implemented: Logistic Regression and Random Forest Classifier
- → Logistic Regression: binary subset (2 - very happy vs. 0 - not happy), SMOTE for class balance
- → Random Forest: binary target (is_very_happy), class_weight='balanced', engineered features (age × health, years_married)
- → Validation: 80/20 train-test split; Random Forest also evaluated with 10-fold cross-validation

SUPERVISED MODELING RESULTS

Logistic Regression

Class	Precision	Recall	F1-Score
Not too Happy (0) 😐	0.67	0.77	0.72
Very Happy (1) ☺	0.89	0.83	0.86
Accuracy		0.81	
Macro Avg	0.78	0.80	0.79
Weighted Avg	0.82	0.81	0.82

SUPERVISED MODELING INTERPRETATION

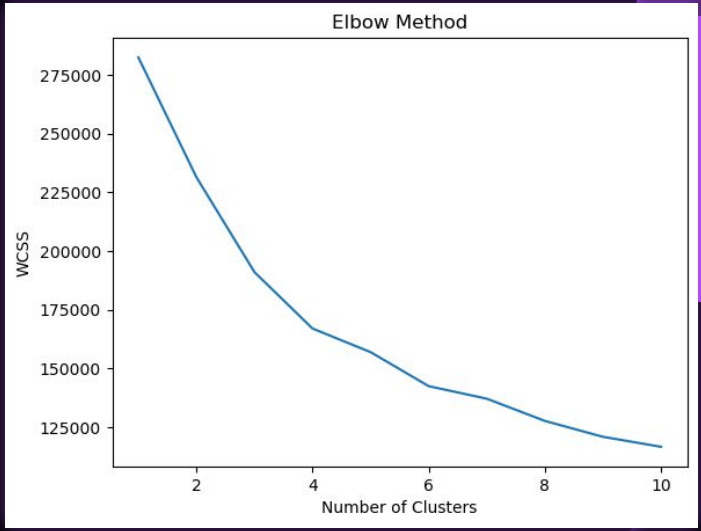


- Our model reveals that physical well-being and social connections (Marriage) are far more predictive of being very happy than traditional demographic markers like age or the specific year the survey was taken.
- Being a woman also has a positive impact on being very happy

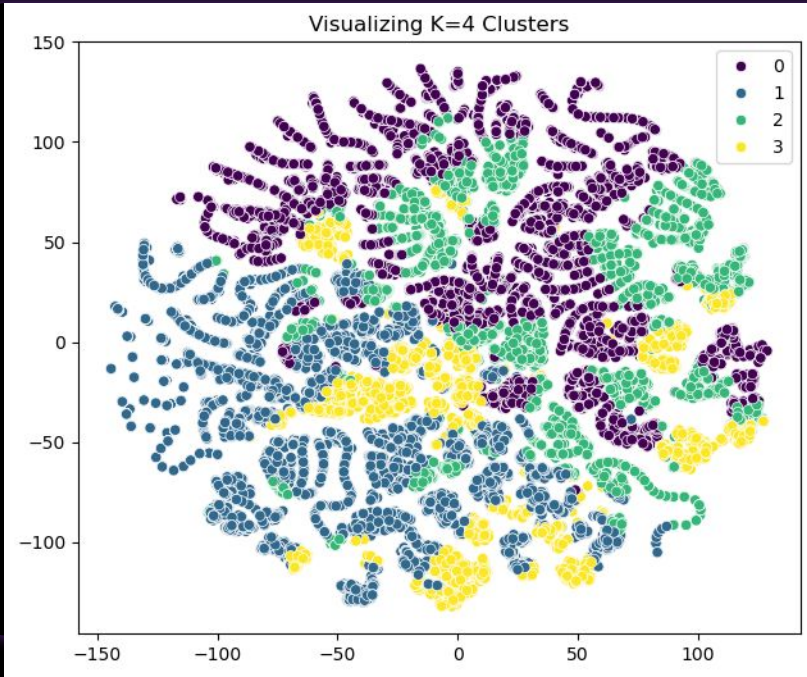
UNSUPERVISED MODELING APPROACH

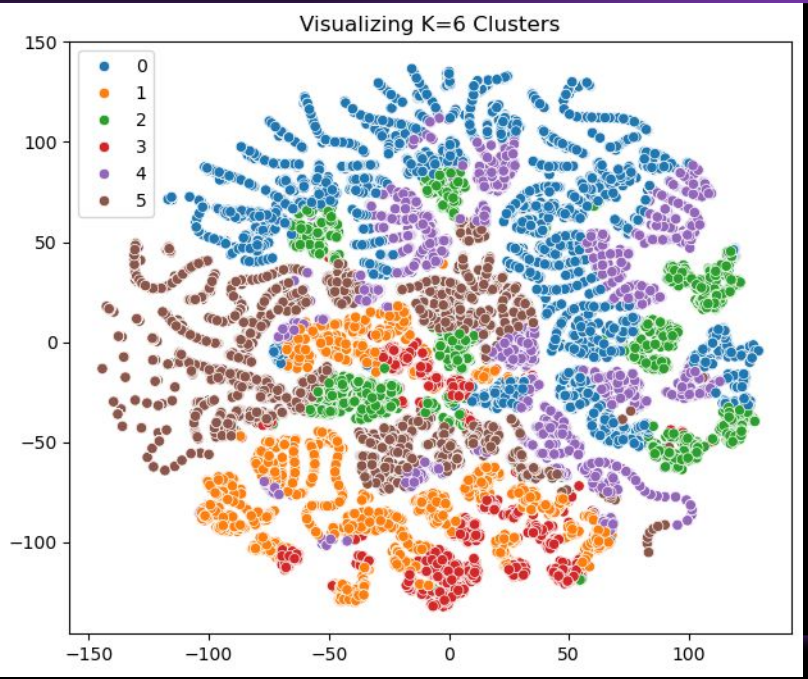
K-Means Clustering

- ⟶ Goal: do respondents naturally group into distinct wellbeing profiles?
- ⟶ Features: happy, health, life excitement, education, age, age×health interaction, years married
- ⟶ K selection: elbow method tested K=3, 4, 6 → elbow visible at K=4 and K=6
- ⟶ Silhouette scores: K=3 (0.181), K=4 (0.184), K=6 (0.186) — all low, marginal differences
- ⟶ K=4 selected: better Davies-Bouldin Index (1.575 vs 1.608 for K=6), more interpretable



UNSUPERVISED MODELING RESULTS





K-Means Clustering Averages

Goal: Do respondents naturally group into distinct wellbeing profiles?

Features: happy, health, life excitement, education, age, age×health interaction, years married

- Cluster with highest happiness
  - Mid 30's
  - Married for 4 years on average
  - Average of 14 years of education (2 yrs of college)
  - High health
  - Averagely exciting life

ETHICS AND LIMITATIONS

- Binary sex variable
- Self-report bias
- Correlation, not causation
- Lots of NAs
- GSS mode shift (2021)
- Non-institutionalized adults only
- Given more time, perform an anomaly detection

CONCLUSION

- Happiness is predictable at ~81% accuracy using GSS data
- Strongest predictors: self-rated health, marital status, sex (Female vs Male), and education
- The gender happiness gap is real and historically documented; women's advantage has narrowed or reversed over 50 years
- The data does not naturally cluster, reinforcing that happiness is a continuous, multidimensional outcome
- Future work: incorporate post-2021 gender identity variables; apply anomaly detection to surface unexpected wellbeing profiles

THANK YOU

DEMO LINK HERE: https://women-ml.streamlit.app/